

BHADRINATHAN A

Backend & Full-Stack Engineer | Technical Lead

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PROFESSIONAL SUMMARY

Backend and full-stack engineer with four years across internship and full-time engineering, progressing to Technical Lead at BloomSkillTech. Own architecture, API and data-model design, implementation and deployment across Python/FastAPI, React and containerized infrastructure. Published two FastAPI libraries; built real-time platforms and Rust security tooling. Hands-on MERN experience, team CTF wins and training delivered to 1,000+ students across colleges. Expanding into AI retrieval and agent workflows.

TECHNICAL SKILLS

Backend: Python, FastAPI, SQLAlchemy, Go, Rust, Axum, REST APIs, SSE, WebSockets, Celery

Web & mobile: JavaScript, TypeScript, React, Next.js, MongoDB, Express, Node.js (MERN), React Native

Data & messaging: PostgreSQL, Redis/Valkey, TimescaleDB, Alembic, MinIO/S3-compatible storage, pub/sub

Infrastructure & quality: Linux, Docker, WireGuard, DNS, Traefik/Nginx, GitHub Actions, CI/CD, Pytest

Security & systems: OAuth, access control, Linux sandboxing, filesystem forensics, CTF problem-solving

AI learning & applied exploration: RAG, pgvector, LangChain, LangGraph, MCP, LLM APIs

PROFESSIONAL EXPERIENCE

BloomSkillTech | Technical Lead - EdTech Platform

Jan 2025 - Present | Salem, Tamil Nadu (Hybrid)

- Led the end-to-end build of a two-sided marketplace connecting trainers and institutions, taking requirements through architecture, data modelling, implementation and production launch.
- Directed engineering across React/Next.js interfaces, FastAPI/PostgreSQL services and cloud infrastructure; owned the technical roadmap and tradeoffs in delivery, infrastructure cost and scalability.
- Established CI/CD pipelines, automated testing, code review and phased delivery so features could move through repeatable validation and release workflows.
- Combined hands-on development with engineering guidance, coordinating frontend, backend and deployment work across the product lifecycle.

BloomSkillTech | Software Developer

May 2023 - Dec 2024 | Salem, Tamil Nadu (Hybrid)

- Built the full stack of a cloud labs platform: browser-accessible interfaces, backend APIs and environment-provisioning services, enabling development sessions without local setup.
- Implemented containerized lab environments for reproducible sessions and a WireGuard VPN layer providing private, isolated access to each lab.
- Built orchestration to provision, scale and tear down environments automatically, managing the session lifecycle and reducing unnecessary idle infrastructure.

Cyber Crime Police Station, Salem | Software Engineer Intern

Aug 2022 - Mar 2023 | Salem, Tamil Nadu

- Sole developer of a crime-records CRM with case timelines, record search and investigation tracking, allowing officers to log and monitor cases through one interface.
- Implemented the React frontend, Go backend APIs and PostgreSQL data layer; containerized and deployed the application with Docker.

ACHIEVEMENTS & TRAINING

TOM CTF: Member of the 1st-place team. **YUKTHI CTF:** Member of the 3rd-place team, conducted by Selfmade Ninja Academy.

Technical training: Trained 1,000+ students from multiple colleges, bringing practical software development experience into teaching and mentorship.

SELECTED ENGINEERING PROJECTS & OPEN SOURCE

fastapi-querybuilder | Python, FastAPI, SQLAlchemy

- Published an MIT-licensed library that converts URL query parameters into SQLAlchemy expressions, with 14 comparison operators, nested relationship joins, recursive search, sorting, pagination and soft-delete support.

fastapi_sse_events | FastAPI, Redis, SSE

- Published a decorator-based event-streaming library; Redis pub/sub distributes events across workers so real-time updates can operate across multi-process deployments.

ZeroCode | Rust, Axum, PostgreSQL, Linux

- Built a 7-crate service for executing untrusted code across 20 languages. Applied namespaces, read-only rootfs, cgroup v2, seccomp BPF, Landlock and capability restrictions; exercised isolation with 130+ adversarial tests.
- Used PostgreSQL LISTEN/NOTIFY and SKIP LOCKED for job dispatch and competing workers, with SSE for live output streaming.

ZeroVPN | Rust, Axum, React, WireGuard, WebAssembly

- Built a VPN management platform with peer control, traffic statistics, DNS and authentication; shared a Rust wire schema with the browser through WebAssembly and MessagePack over WebSocket.
- Organized authentication, WireGuard control, statistics and DNS into a multi-crate backend; added TOTP, sessions and API tokens, with startup checks rejecting placeholder secrets.

NullVeil | Rust, Filesystems, Digital Forensics

- Built read-only recovery tooling with hand-written FAT, exFAT, NTFS, ext and APFS parsers, signature carving, SHA-256 manifests and an interactive terminal UI; kept write operations outside the source-disk reader.
- Implemented NTFS MFT traversal and APFS B+-tree parsing, plus deduplication, verification re-reads and a JSON-lines audit trail for recovered artifacts.

AlgoTrade | FastAPI, React, Rust, Celery, Redis, TimescaleDB

- Built a 16-section dashboard and trading workflows with a Rust risk engine. Fanned broker WebSocket ticks through Redis to strategy loops and the browser, stored ticks in TimescaleDB, and ran background tasks through Celery.
- Implemented position sizing, options Greeks and per-tick risk checks in Rust; supported paper-trading fills and strategy supervision through Celery workers.

Voxiloud | React Native, Recoil, NativeWind

- Built a text-to-speech mobile app with speed and volume controls, translation, file-to-text and searchable saved history; managed global state with Recoil.

Docker Stats Monitor | Python, Docker, WebSockets

- Built a live dashboard streaming container status, CPU, memory, network and block-I/O metrics over WebSockets for ongoing runtime visibility.

EDUCATION & ACADEMIC PROJECT

B.E. Electronics and Communication Engineering | 2019 - 2023

Muthayammal Engineering College, Rasipuram, Tamil Nadu | First Class

Final-year project - Gesture-Controlled Electronics: Built an IoT/electronics prototype to control electronic components through gestures, applying hardware-software integration to an interactive control system.